

Super-Eddington Little Blue Dots Reionize Helium Too Early

JWST's faint AGN may over-ionize helium unless only a small fraction accrete at extreme rates.

arXiv:2608.28745 — Super-Eddington Little Blue Dots May Reionize Helium Too Early

arXiv:2608.28745 doi:10.48550/arXiv.2608.28745



Background

JWST found abundant faint AGN at $z \gtrsim 4$ with broad lines and blue continua, dubbed Little Blue Dots. Some models link them to Little Red Dots via super-Eddington accretion and viewing angle.

Can super-Eddington LBDs reionize helium without violating the observed end at $z \approx 3$?

What this paper is selling

The authors compute helium reionization histories for super-Eddington spectra. They find that if LBDs dominate the ionizing budget, at most $\sim 10\%$ can be highly super-Eddington.

Key results to remember

- Helium reionization ending no earlier than $z \approx 3$ limits super-Eddington LBDs to $\lesssim 10\%$.
- Majority super-Eddington requires escape fractions $\lesssim 1.5\%$ or lower abundance.
- Constraints depend strongly on the extreme-UV spectra of super-Eddington black holes.

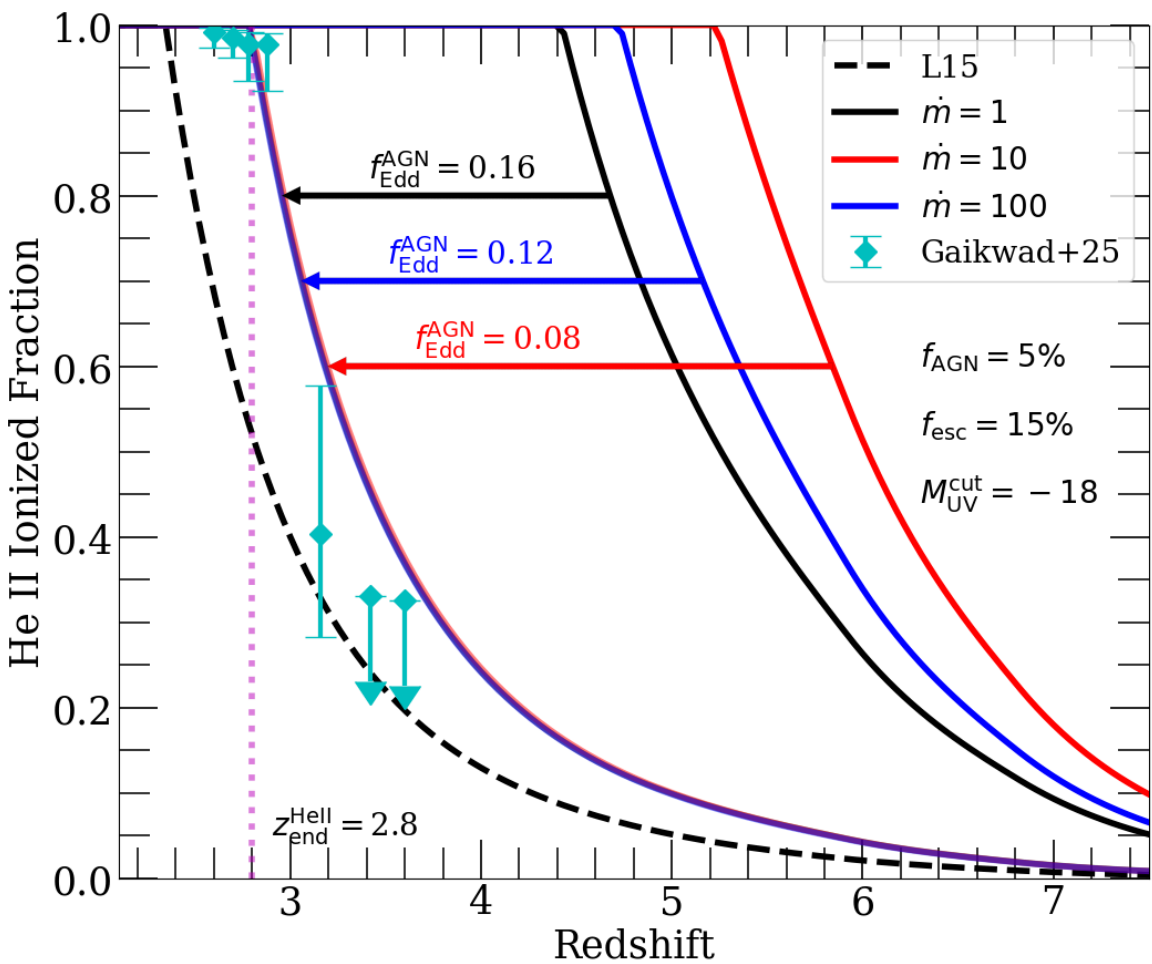


Figure 3 | Helium reionization histories for different SEDs show that completing reionization by $z=2.8$ requires only a small super-Eddington fraction.

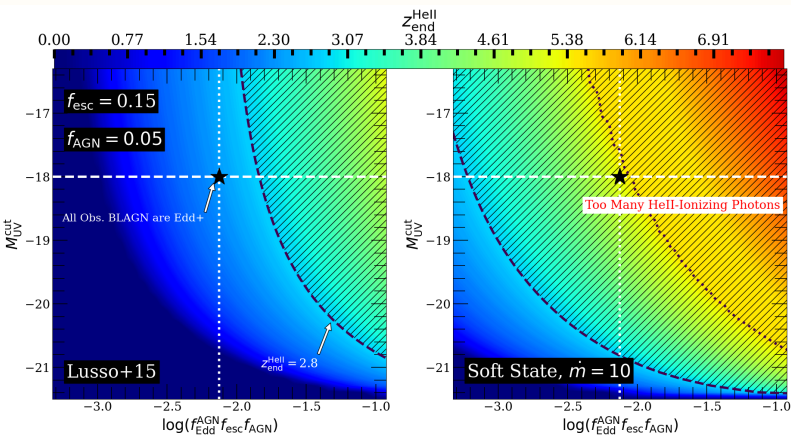


Figure 4 | Excluded parameter space: the hatched region yields helium reionization ending too early, disfavoring large super-Eddington fractions.